UNIVERSITY OF SCIENCE

FACULTY OF INFORMATION TECHNOLOGY

# Lab 02: Linear Regression Project

**Introduction to Machine Learning**


***Sinh viên thực hiện:***

***Nguyễn Thành Đạt - 23127343***

***Trần Quang Đạo - 21127498***

***Trịnh Hạnh - 21127040***

***Nguyễn Cát Huy - 21127300***

***Nguyễn Tuấn Tuấn - 21127562***

***Giảng viên hướng dẫn:***

***Thầy Bùi Tiến Lên***

***Thầy Võ Nhật Tân***

***Thầy Lê Nhựt Nam***


**Ho Chi Minh, 2026**

**Table 1:** Detailed Team Contribution and Responsibilities

| No. | Full Name | Student ID | Tasks Performed | Contribution |
|---|---|---|---|---|
| 1 | Nguyễn Thành Đạt | 23127343 | • Overall model design<br>• Implemented Model 2 & Model 7<br>• Data Visualization<br>• Conducted EDA<br>• Wrote report | 100% |
| 2 | Trần Quang Đạo | 21127498 | • Implemented Model 1 & Model 5<br>• Data preprocessing<br>• Data Visualization | 100% |
| 3 | Trịnh Hạnh | 21127040 | • Implemented Model 4 & Model 6<br>• Data Visualization<br>• Deployed Streamlit Application | 100% |
| 4 | Nguyễn Cát Huy | 21127300 | • Implemented Model 3<br>• Dataset research & exploration | 80% |
| 5 | Nguyễn Tuấn Tuấn | 21127562 | No contribution | 0% |

# Contents

**1 Problem description and motivation** **3**
1.1 Problem Description . . . 3
1.2 Motivation . . . 3

**2 Dataset description** **3**
2.1 Overview and Source . . . 3
2.2 Feature Definitions . . . 4
2.3 Statistical Summary . . . 4

**3 Data preprocessing and visualization** **5**
3.1 Data Visualization Insights . . . 5
3.2 Data Preprocessing . . . 8

**4 Model Design and Explanation** **9**
4.1 Mathematical Foundation . . . 9
4.2 The Evolution of 7 Model Architectures . . . 9

**5 Evaluation and Comparison of Results** **10**
5.1 Evaluation Metrics . . . 10
5.2 Comparative Performance Analysis . . . 11
5.3 Key Findings and Discussion . . . 11

**6 Interpretation and Insights Drawn from the Dataset** **11**
6.1 Principal Drivers of Housing Value . . . 11
6.2 The Impact of Feature Engineering . . . 12
6.3 Data Anomalies and Limitations . . . 12
6.4 Socio-Economic Summary . . . 12
6.5 Quantitative Feature Importance . . . 12

**7 Brief Description of the Application** **13**
7.1 Application Overview and Motivation . . . 13
7.2 User Interface (UI) Design . . . 15
7.3 Input and Output Functionality . . . 15
7.4 Model Integration and Pipeline Logic . . . 15

**References** **17**

# 1 Problem description and motivation

## 1.1 Problem Description

Real estate valuation is a complex process influenced by a multitude of factors, ranging from physical attributes of a property to its geographical location and the socio-economic status of the surrounding neighborhood. In this project, we address the challenge of predicting the Median House Value in various California districts using the California Housing dataset.

The core of this problem is a Supervised Learning - Regression task. Given a set of input features such as median_income, housing_median_age, total_rooms, and geographical coordinates (longitude, latitude), the goal is to build a predictive model that can estimate the market value of a home with high precision. This requires the model to capture not only the linear relationships between features but also the spatial patterns inherent in geographical data.

## 1.2 Motivation

The motivation behind this project stems from both practical and technical perspectives:

- **Practical Utility:** Accurate housing price predictions are crucial for various stakeholders. Homeowners need to know the fair value of their assets, real estate investors require data-driven insights to identify undervalued properties, and financial institutions rely on these estimates for mortgage lending and risk assessment.
- **Technical Exploration:** From a Machine Learning standpoint, the California Housing dataset presents fascinating challenges:
  - **Multicollinearity:** Dealing with highly correlated variables (e.g., total rooms vs. total bedrooms).
  - **Spatial Analysis:** Exploring how location (coastal vs. inland) impacts value, which led us to implement Spatial Clustering (K-Means).
  - **Model Optimization:** The drive to go beyond a simple "Baseline" model by experimenting with 7 different model iterations (Log-transformations, Interaction features, VIF filtering) to achieve the best $R^2$ score.
- **End-to-End Integration:** We were motivated to bridge the gap between a static data science experiment and a functional product. By deploying the final model as a Streamlit Web Application, we demonstrate how complex ML models can be made accessible and intuitive for end-users.

# 2 Dataset description

## 2.1 Overview and Source

The dataset used in this project is the California Housing Dataset, which was originally derived from the 1990 U.S. Census. It contains information regarding the housing stock in a given California district (block group). A

block group is the smallest geographical unit for which the U.S. Census Bureau publishes sample data, typically containing a population of 600 to 3,000 people.

The dataset consists of 20,640 observations (records), each representing a specific district, with 10 initial attributes (features) including 9 numerical variables and 1 categorical variable.

## 2.2 Feature Definitions

Understanding the features is crucial for interpreting the model's predictions. The attributes are categorized as follows:

- **Geographical Features:**
  - `longitude`: A measure of how far west a house is; a higher value is farther west.
  - `latitude`: A measure of how far north a house is; a higher value is farther north.
- **Property Characteristics:**
  - `housing_median_age`: Median age of a house within a block; a lower number is a newer building.
  - `total_rooms`: Total number of rooms within a block.
  - `total_bedrooms`: Total number of bedrooms within a block.
- **Demographic Features:**
  - `population`: Total number of people residing within a block.
  - `households`: Total number of households, a group of people residing within a home unit, for a block.
- **Economic Features:**
  - `median_income`: Median income for households within a block of houses (measured in tens of thousands of US Dollars).
  - `median_house_value (Target Variable)`:Median house value for households within a block (measured in US Dollars).
- **Location Feature:**
  - `ocean_proximity`: Location of the house w.r.t ocean/sea (Categorical: `<1H OCEAN, INLAND, ISLAND, NEAR BAY, NEAR OCEAN)`.

## 2.3 Statistical Summary

Below is the descriptive statistics summary of the numerical features in the dataset. This summary provides insights into the scale, central tendency, and dispersion of the data.

**Table 2:** Statistical Summary Housing Dataset

| Feature | Count | Mean | Std | Min | 25% | 50% | 75% | Max |
|---|---|---|---|---|---|---|---|---|
| Longitude | 20,640 | -119.57 | 2.00 | -124.35 | -121.80 | -118.49 | -118.01 | -114.31 |
| Latitude | 20,640 | 35.63 | 2.14 | 32.54 | 33.93 | 34.26 | 37.71 | 41.95 |
| Housing Median Age | 20,640 | 28.64 | 12.59 | 1.00 | 18.00 | 29.00 | 37.00 | 52.00 |
| Total Rooms | 20,640 | 2,635.76 | 2,181.62 | 2.00 | 1,447.75 | 2,127.00 | 3,148.00 | 39,320.00 |
| Total Bedrooms | 20,433 | 537.87 | 421.39 | 1.00 | 296.00 | 435.00 | 647.00 | 6,445.00 |
| Population | 20,640 | 1,425.47 | 1,132.46 | 3.00 | 787.00 | 1,166.00 | 1,725.00 | 35,682.00 |
| Households | 20,640 | 499.54 | 382.33 | 1.00 | 280.00 | 409.00 | 605.00 | 6,082.00 |
| Median Income | 20,640 | 3.87 | 1.90 | 0.50 | 2.56 | 3.53 | 4.74 | 15.00 |
| Median House Value | 20,640 | 206,855.82 | 115,395.62 | 14,999.00 | 119,600.00 | 179,700.00 | 264,725.00 | 500,001.00 |

**Key Observations and Data Quality Insights:**

1. **Missing Values Identification:** The count for `total_bedrooms` is 20,433, which is less than the total entries (20,640). This confirms that approximately 1% of the data is missing for this feature, necessitating an imputation strategy in the next phase

2. **Data Capping (Outlier Alert):** Both `media_income` and `median_house_value` show clear evidence of being capped. The maximum house value is exactly $500,001$ and the maximum income is 15.0. This artificial limit will impact the model's accuracy for high-end properties.

3. **Scale Disparity:** There is a vast difference in scales between features. longitude is around -120, while `total_rooms` goes up to 39,320. This indicates that Feature Scaling (Normalization/Standardization) is mandatory for algorithms like Linear Regression to perform optimally.

4. **Skewness:** For features like `total_rooms`, population, and households, the Mean is significantly higher than the Median (50%). This suggests that the distributions are right-skewed with long tails of high-value outliers.

# 3 Data preprocessing and visualization

## 3.1 Data Visualization Insights

Before building the models, we conducted an extensive Exploratory Data Analysis (EDA) to understand the underlying patterns and anomalies in the California housing market.

- **Target Variable Distribution:** The Histogram of `median_house_value shows` a significant spike at $500,001$. This indicates that prices above this threshold were capped during data collection, which may introduce bias in our regression models.

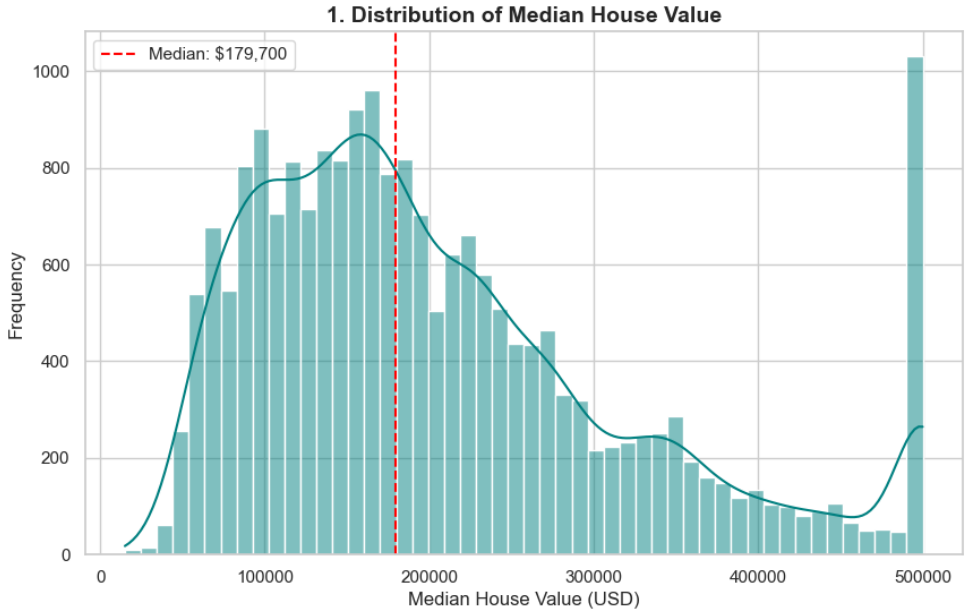


**Figure 1:** Distribution of Median House Value

- **Correlation Analysis:** The Heatmap reveals that `median_income` has the strongest positive correlation (0.69) with house prices. Conversely, latitude and longitude show a strong negative correlation, reflecting the geographical shape of California.

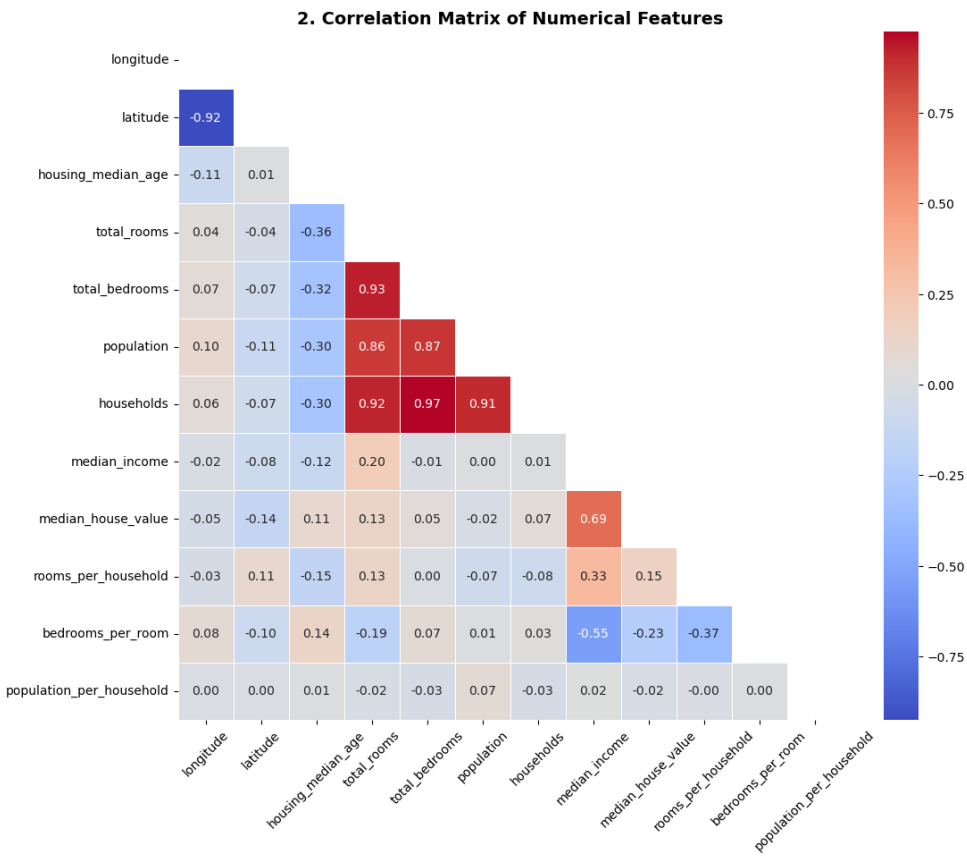


**Figure 2:** Correlation Matrix of Numerical Features

- **Geographical Trends:** The Scatter Map (Longitude vs. Latitude) clearly shows that houses located near the coast (San Francisco and Los Angeles areas) have much higher values compared to inland regions.

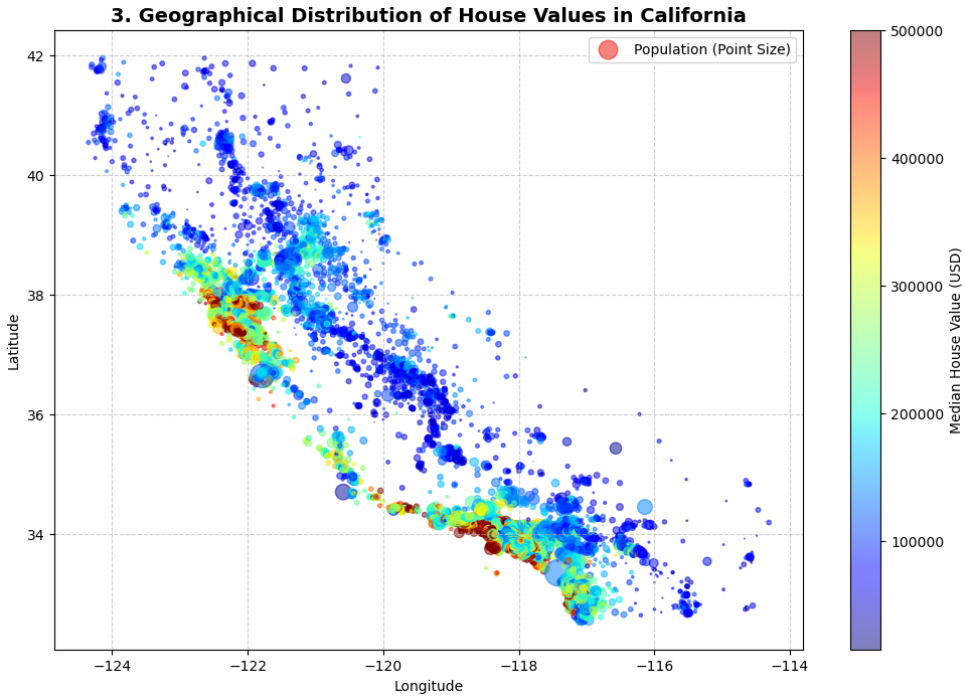


**Figure 3:** Geographical Distribution of House Values in California

- **Linearity Outliers:** The Regression Plot confirms a linear relationship between income and price. However, the Boxplots for variables like population and `total_rooms` show many extreme outliers, which we addressed during preprocessing.

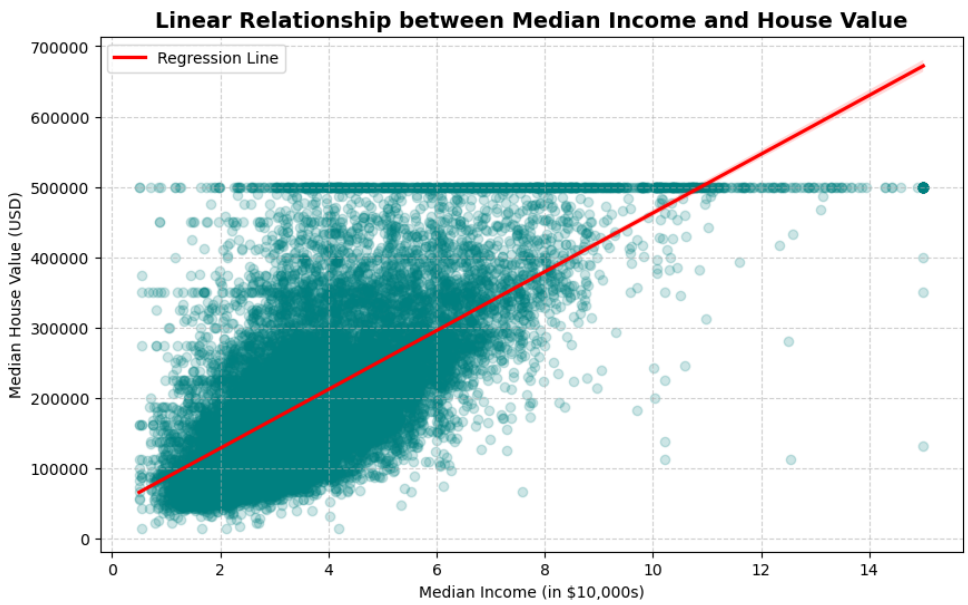


**Figure 4:** Geographical Distribution of House Values in California

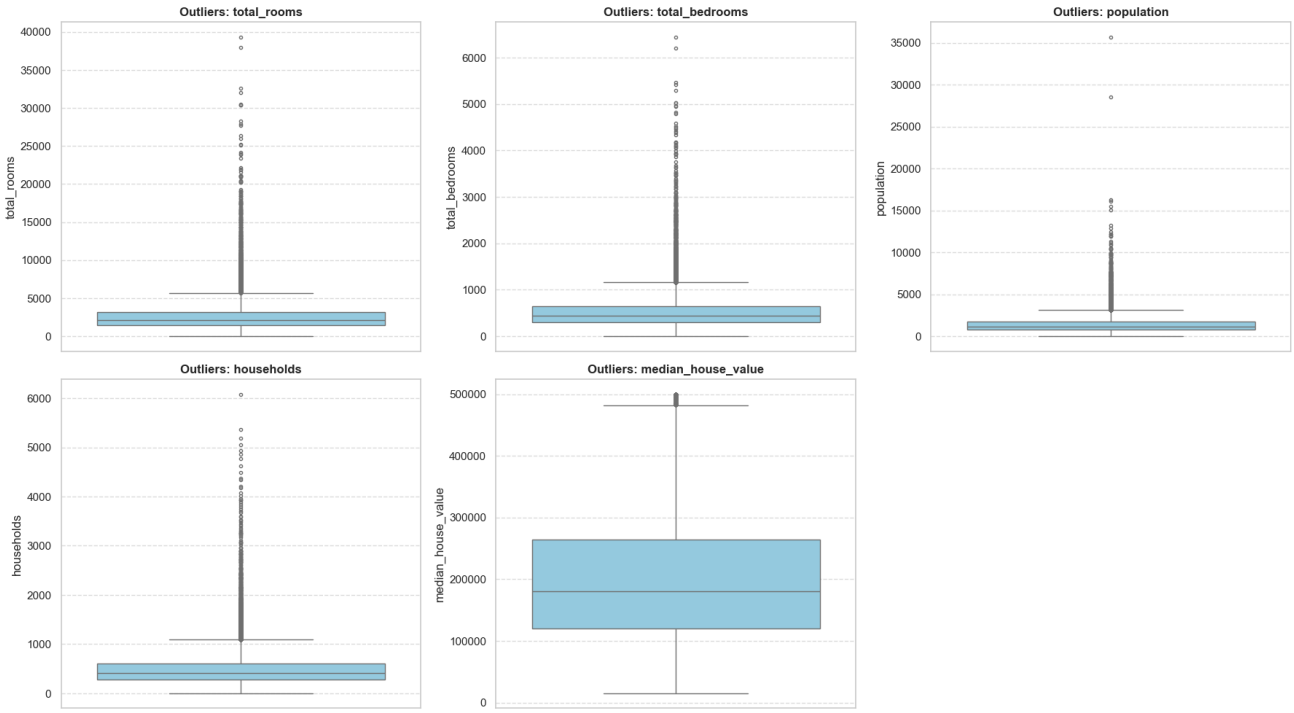


**Figure 5:** Boxplots

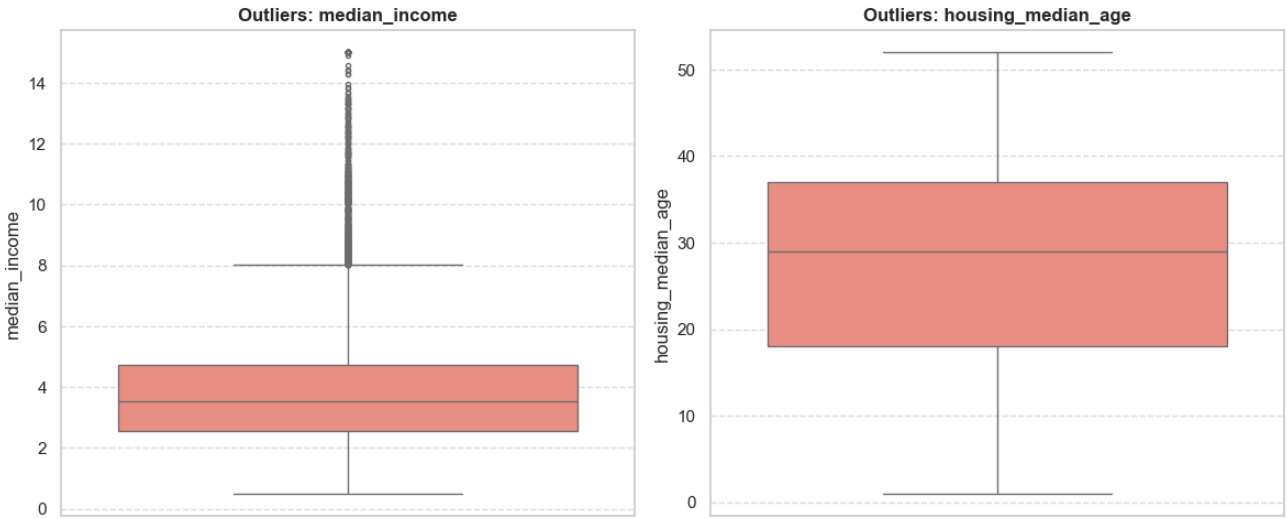


**Figure 6:** Boxplots

## 3.2 Data Preprocessing

To prepare the data for Linear Regression, we implemented a robust pipeline:

1. **Handling Missing Values:** We identified 207 missing entries in `total_bedrooms`. We used the Median Imputation strategy to fill these gaps, ensuring no data was lost while maintaining the feature's distribution.

2. **Feature Engineering: Raw counts (like total rooms) are less informative than ratios. We created three new synthetic features:**

   - `rooms_per_household`: Average rooms per house.

- `bedrooms_per_room`: Ratio of bedrooms to total rooms.
- `population_per_household`: Average number of residents per house.

3. **Categorical Encoding:** The `ocean_proximity` feature was transformed into 5 binary columns using One-Hot Encoding to make it compatible with the Linear Regression algorithm.
4. **Spatial Clustering:** Recognizing that location is a key driver, we applied K-Means Clustering on `longitude` and `latitude`. This converted complex coordinates into 5 distinct geographic "Clusters," allowing Model 6 to capture regional price variations effectively.

# 4 Model Design and Explanation

## 4.1 Mathematical Foundation

The foundation of our project is the Ordinary Least Squares (OLS) Regression. The fundamental goal is to find the vector of coefficients $\beta$ that minimizes the Sum of Squared Residuals (SSR):

$$RSS = \sum_{i=1}^{n}(y_i - \hat{y}_i)^2 = \sum_{i=1}^{n}(y_i - (\beta_0 + \sum_{j=1}^{p}\beta_j x_{ij}))^2$$

However, a raw OLS approach often fails in real estate data due to non-normality and multicollinearity. Therefore, we designed a Multi-Stage Modeling Pipeline to transition from a naive baseline to a sophisticated spatial-aware regressor.

## 4.2 The Evolution of 7 Model Architectures

**Stage 1: Handling Data Redundancy**

- **Model 1 (Baseline):** A standard OLS model using raw features to establish a performance floor.
- **Model 2 (VIF Optimization):** We addressed the inflation of coefficient variance caused by Multicollinearity. We calculated the Variance Inflation Factor (VIF) for each feature $j$:

$$VIF_j = \frac{1}{1 - R_j^2}$$

  Features with $VIF > 10$ (such as `total_rooms` and `households`) were analyzed and redundant variables were removed to ensure model stability.

**Stage 2: Structural Transformations (Model 3  Model 5)**

- **Model 3 (Log-Log Regression):** Real estate data often exhibits exponential growth patterns. By applying a natural logarithm transformation to both $X$ and $Y$, we modeled elasticity:

$$\ln(Y) = \beta_0 + \beta_1 \ln(X_1) + \cdots + \epsilon$$

  This effectively normalized the "heavy-tail" distributions observed in the EDA phase

- **Model 5 (Polynomial Expansion):** To capture non-linear diminishing returns (e.g., the non-linear impact of housing$_m edian_a ge), weintroducedsecond - degreeterms : y = \beta_0 + \beta_1 x + \beta_2 x^2 + \epsilon$

**Stage 3: Feature Engineering and Spatial Intelligence (Model 4 Model 6)**

- **Model 4 (Interaction Ratios):** We engineered socio-economic indicators to move beyond raw counts:

$$\text{rooms_per_household} = \frac{\text{total_rooms}}{\text{households}}$$

- **Model 6 (Spatial Clustering via K-Means) - The Champion Model:** Standard Linear Regression treats coordinates as independent linear variables, which fails to capture complex geography. We integrated Unsupervised Learning by clustering longitude and latitude into $K = 5$ zones. The Cluster labels were converted into One-Hot Encoded dummy variables. This allowed the model to learn region-specific intercepts:

$$\hat{y} = \beta_0 + \sum \beta_{cluster_i} D_i + \sum \beta_j x_j$$

  This hybrid approach effectively localized the regression, significantly boosting the $R^2$ score.

**Robustness Analysis (M7):**

- **Model 7 (Outlier Sensitivity):** We truncated the dataset to remove the $500,001$ price cap and extreme income outliers. This served as a sensitivity test to understand how the "luxury" segment influences the general linear trend.

# 5 Evaluation and Comparison of Results

## 5.1 Evaluation Metrics

To rigorously assess the performance of our 7 models, we utilized three standard regression metrics. These metrics provide a comprehensive view of the model's accuracy and its ability to generalize to unseen data:

- **Mean Absolute Error (MAE):** Represents the average magnitude of the errors in a set of predictions, without considering their direction.

$$MAE = \frac{1}{n} \sum_{i=1}^{n} |y_i - \hat{y}_i|$$

- **Root Mean Squared Error (RMSE):** The square root of the average of squared differences between prediction and actual observation. It penalizes larger errors more heavily than MAE.

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^{n} (y_i - \hat{y}_i)^2}$$

- **Coefficient of Determination ($R^2$ Score):** Indicates the proportion of the variance for the dependent variable that's explained by the independent variables in the model.

$$R^2 = 1 - \frac{SS_{res}}{SS_{tot}}$$

### 5.2 Comparative Performance Analysis

The table below summarizes the performance of all 7 iterations on the test dataset:

**Table 3:** Model Performance Comparison

| Model | MAE | RMSE | R2 |
|---|---|---|---|
| Model 6: Spatial Cluster | 49,924.74 | 71,084.90 | 0.6144 |
| Model 4: Interaction | 50,787.82 | 72,565.19 | 0.5982 |
| Model 1: Baseline | 50,888.66 | 72,668.54 | 0.5970 |
| Model 5: Polynomial | 51,565.22 | 74,201.57 | 0.5798 |
| Model 2: VIF Optimized | 53,818.61 | 77,307.85 | 0.5439 |
| Model 7: Outlier Cleaned | 52,449.50 | 81,337.26 | 0.4951 |
| Model 3: Log-Log | 60,254.91 | 685,760.87 | -34.8871 |

### 5.3 Key Findings and Discussion

1. **Superiority of Spatial Awareness:** Model 6 emerged as the best performer with an $R^2$ of 0.614. By integrating geographical clusters, the model effectively captured the "location premium" of California's coastal regions, which a global linear model (M1) failed to do.

2. **Impact of Feature Engineering:** The transition from Model 1 to Model 4 shows that creating ratios (e.g., `rooms_per_household`) provides better economic signals than raw counts, leading to a noticeable drop in MAE.

3. **The Log-Log Paradox (Model3):** While Model 3 helped in normalizing data distributions, the extremely high RMSE and negative $R^2$ on the original scale indicate that the linear relationship in log-space did not translate perfectly back to the original price scale without further non-linear adjustments.

4. **Resilience to Outliers (Model 7):** Removing outliers in M7 slightly lowered the $R^2$ compared to M6. This suggests that while outliers introduce noise, they also contain high-value information that the spatial clustering in M6 handles more gracefully than simple deletion.

## 6 Interpretation and Insights Drawn from the Dataset

### 6.1 Principal Drivers of Housing Value

The multi-model analysis confirms that housing prices in California are not randomly distributed but follow a strict socio-economic and geographical hierarchy.

- **Income Elasticity:** `median_income` remains the most dominant predictor across all 7 models. The high correlation (0.69) suggests that for every unit increase in the median income of a district, the house value rises significantly. This validates that the purchasing power of residents is the primary floor for real estate pricing.

- **The "Location Premium":** Our spatial clustering (Model 6) revealed that proximity to the coast is a "multiplier" for value. A house in a coastal cluster (e.g., near San Francisco) with the same physical attributes as one in the Central Valley can be priced 2-3 times higher.

## 6.2 The Impact of Feature Engineering

The superiority of Model 4 and Model 6 over the Baseline (Model 1) underscores the importance of domain-specific feature engineering:

- **Density vs. Quantity:** Raw counts like `total_rooms` proved to be less impactful than the derived ratio `rooms_per_household`. This suggests that the "spaciousness" of individual homes in a district is a better proxy for luxury and value than the total number of rooms in the entire block.
- **Geographical Localization:** Linear regression traditionally assumes a global relationship. However, by introducing K-Means Clusters, we forced the model to recognize local market trends. This insight proves that the California market is fragmented into distinct sub-markets that require localized intercepts.

## 6.3 Data Anomalies and Limitations

During our analysis, we identified a critical data anomaly: The $500,001$ Price Cap.

- A significant portion of the data is clustered exactly at $500,001$. This is a clear case of "censored data" where values above the threshold were not recorded accurately.
- **Insight for Modeling:** This cap introduces a systematic bias, as the model tries to find a linear fit for data that is artificially flattened at the high end. This explains why the error (RMSE) remains relatively high; the model cannot accurately learn the features of "ultra-luxury" homes because their true values are hidden.

## 6.4 Socio-Economic Summary

In conclusion, the dataset paints a picture of a highly polarized market. Wealth is concentrated in specific coastal clusters where high income and limited housing supply (reflected in higher `housing_median_age`) drive prices to the maximum threshold. For future improvements, incorporating external data such as school ratings or crime rates could potentially explain the remaining variance that census data alone cannot capture.

## 6.5 Quantitative Feature Importance

The importance of each feature is quantified through the **Regression Coefficients** of our best-performing model (Model 6). These coefficients provide a mathematical representation of how each variable scales the predicted house value:

**Table 4:** Variance Inflation Factor (VIF) Results

| Feature | VIF |
|---|---|
| Longitude | 928.60 |
| Latitude | 827.93 |
| Housing Median Age | 8.14 |
| Total Rooms | 40.73 |
| Total Bedrooms | 94.24 |
| Population | 17.49 |
| Households | 93.93 |
| Median Income | 10.09 |

| Feature | VIF |
|---|---|
| Rooms per Household | 9.30 |
| Bedrooms per Room | 25.71 |
| Population per Household | 1.16 |
| Ocean: INLAND | 2.75 |
| Ocean: ISLAND | 1.00 |
| Ocean: NEAR BAY | 1.68 |
| Ocean: NEAR OCEAN | 1.31 |

- **Median Income Impact:** The coefficient for `median_income` is approximately $39,122$. This implies that, holding all other factors constant, a one-unit increase in the median income category corresponds to an average increase of over $\$39,000$ in property value.
- **Structural Density:** The variable `bedrooms_per_room` shows a significant negative coefficient (approx. $-137,358$). This indicates that districts with a higher density of bedrooms relative to total rooms (suggesting smaller, more cramped living spaces) experience a substantial decrease in market valuation.
- **Age Factor:** The positive coefficient for `housing_median_age` (approx. $1,114$) suggests that in the California market, older, well-established neighborhoods often retain or increase in value, possibly due to superior locations or historical prestige.

# 7 Brief Description of the Application

## 7.1 Application Overview and Motivation

To bridge the gap between academic research and practical utility, we developed a web-based application using the Streamlit framework. The primary goal is to provide a user-friendly interface where non-technical stakeholders (e.g., real estate agents, home buyers) can interact with our best-performing model (Model 6: Spatial Cluster) to get instant, data-driven house price estimates.

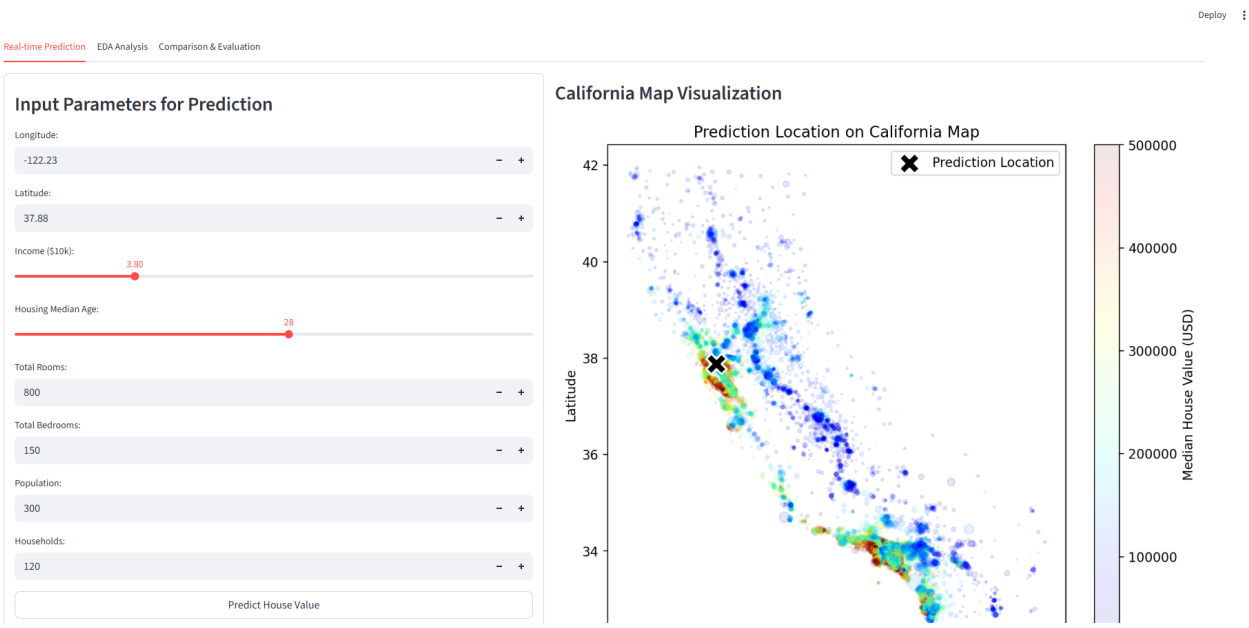


**Figure 7:** Tab1-Real-time Precdiction

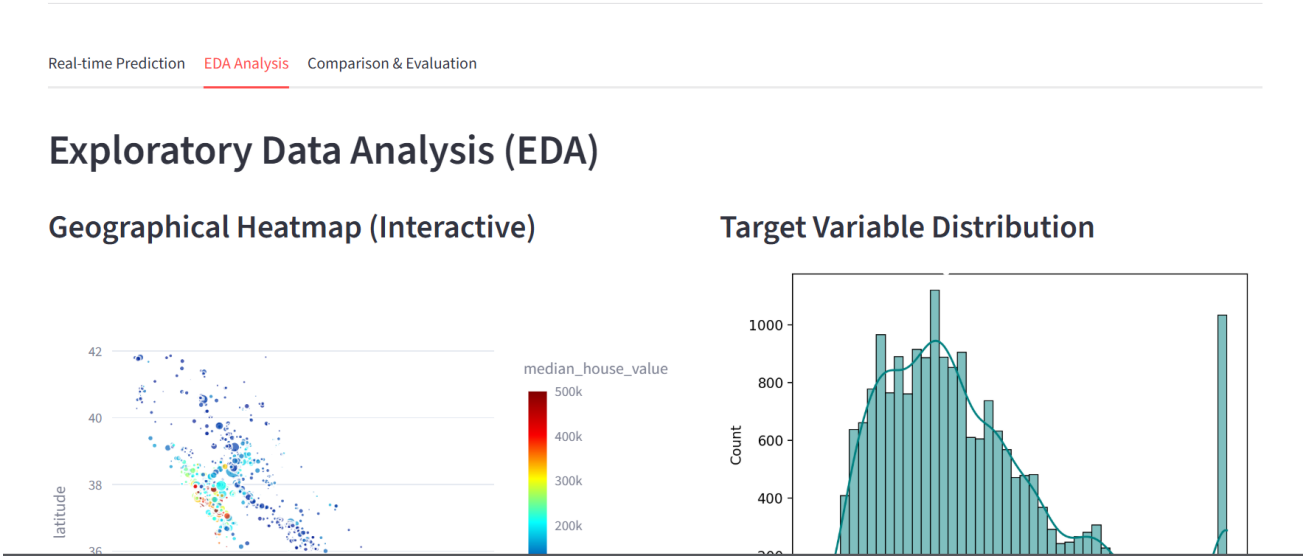


**Figure 8:** Tab2-EDA Analysis

Real-time Prediction EDA Analysis Comparison & Evaluation

## Model Performance Evaluation

| | Model | MAE | RMSE | R2 |
|---|---|---|---|---|
| 0 | M6: Spatial Cluster | 49924.740000 | 71084.900000 | 0.614000 |
| 1 | M4: Interaction | 50787.820000 | 72565.180000 | 0.598000 |
| 2 | M1: Baseline | 50888.660000 | 72668.530000 | 0.597000 |
| 3 | M5: Polynomial | 51565.220000 | 74201.560000 | 0.579000 |
| 4 | M2: VIF Optimized | 53818.610000 | 77307.840000 | 0.543000 |
| 5 | M7: Outlier Cleaned | 52449.490000 | 81337.260000 | 0.495000 |
| 6 | M3: Log-Log | 60254.900000 | 685760.860000 | -34.880000 |

**Figure 9:** Tab3-Comparision Evaluation

## 7.2 User Interface (UI) Design

The application features a minimalist and intuitive design, organized into a three-tab layout to separate different user needs:

- **Tab 1 Real-time Prediction:** The core functional area where users input property features and receive price estimates.
- **Tab 2 Interactive EDA Analysis:** An educational area that displays the key data insights (Correlation Heatmap, Geographical Map, etc.) found during our research.
- **Tab 3 Comparision Evaluation:** A transparency area showing the performance metrics (MAE, RMSE, $R^2$) of all 7 experimental models to justify why Model 6 was deployed.

## 7.3 Input and Output Functionality

The application handles complex data transformations behind a simple I/O interface:

- **Inputs:** Users provide 8 primary features through a combination of st.number_input (for precise coordinates and counts) and st.slider (for income and house age).
- **Visual Feedback:** As a user enters coordinates, an interactive Matplotlib map updates with a high-contrast black 'X' (using zorder=10 and edgecolors='white') to visually confirm the selected geographical location.
- **Output:** Upon clicking the " Predict House Value" button, the estimated `median_house_value` is displayed in a formatted currency box, accompanied by a success animation (st.balloons).

## 7.4 Model Integration and Pipeline Logic

The integration of the trained model into the production environment follows a rigorous pipeline:

1. **Asset Loading:** The app utilizes the joblib library to load two serialized files: model_6.pkl (the regression weights) and spatial_cluster_model.pkl (the K-Means clusterer).

2. **On-the-fly Preprocessing:** Before prediction, the app dynamically transforms the raw 8 inputs into a 20-feature vector:
   - It calculates interaction ratios: `rooms_per_household, bedrooms_per_room, etc.`
   - It uses the loaded K-Means model to assign a Spatial Cluster to the input coordinates.
   - It converts the cluster ID into One-Hot Encoded dummy variables.
3. **Prediction:** The final processed vector is fed into the Linear Regression model to generate the final value.

# References

[1] Aurélien Géron, *Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow: Concepts, Tools, and Techniques to Build Intelligent Systems*, 2nd Edition, O'Reilly Media, 2019. (Tài liệu chính về quy trình xử lý dữ liệu California Housing).

[2] F. Pedregosa *et al.*, "Scikit-learn: Machine Learning in Python," *Journal of Machine Learning Research*, vol. 12, pp. 2825–2830, 2011.

[3] Streamlit, "Streamlit Documentation: The fastest way to build and share data apps," [Online]. Available: `https://docs.streamlit.io/`.

[4] J. D. Hunter, "Matplotlib: A 2D Graphics Environment," *Computing in Science & Engineering*, vol. 9, no. 3, pp. 90–95, 2007.

[5] Michael L. Waskom, "seaborn: statistical data visualization," *Journal of Open Source Software*, vol. 6, no. 60, p. 3021, 2021.

[6] Wes McKinney, "Data Structures for Statistical Computing in Python," *Proceedings of the 9th Python in Science Conference*, 2010, pp. 51–56.

[7] Joblib Development Team, "Joblib: running Python functions as pipeline jobs," [Online]. Available: `https://joblib.readthedocs.io/`.

[8] Kaggle, "California Housing Prices Dataset," [Online]. Available: `https://www.kaggle.com/datasets/shibumohapatra/house-price`. (The primary dataset utilized for model training and performance evaluation in this project).